

ActInf GuestStream 083.1 ~ "Embodied intelligence", Joshua Bongard

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hello and welcome it is July 17th 2024 we're in active inference guest stream 83.1 with Josh bongard thank you for joining this should be very exciting we'll have a presentation and then some discussion so if you're in the live chat please feel free to write any questions and thank you again Josh looking forward to this yeah thank you Daniel and thanks to those of you that are attending online um so my name is Josh bongard I'm a professor of computer science uh here at the University of Vermont and um my bread and butter in my lab is the study of Robotics and Ai and obviously we're in the middle of the uh a current AI summer um so what I wanted to do today is uh show a couple highlights from my group things that I've worked on in the past and that we're working on at the moment um that I hope in the long term will help us realize sort of the long-term vision for a lot of those trying to create intelligent machines which is to create machines that are helpful but but also safe um we're part of the way there but as anyone who's worked with chat GPT or stable diffusion NOS or even has a robot vacuum cleaner at home there's some limitations to our current technology it's hard to create machines that are autonomous and useful and safe all simultaneously so what are the things that we're missing um that's what I wanted to sort of um seed the pool with today and hopefully we can move on to an interactive discussion about it so uh I'm going to leave this slide up and just sort of talk over this slide for a few minutes that'll hopefully generate some food for thought and then questions um this is a snapshot from uh some of the projects I've worked on over the years um first thing you'll probably notice is there's a lot of different robots that have very different structural properties they not only act differently but look very different and that is a fundamental foundation in everything we do uh in my research Group which is to try and understand how the body facilitates uh cognition uh years ago with my PhD advisor we wrote a popular science book called How the body shapes the way we think and we can mean that literally or figuratively um we wrote that book a while ago we made some arguments about how the body shapes the way we think and since that time my group

and others have formulated uh other uh Arguments for why or how the body shapes the way we think and I'm hoping to survey some of those uh today so as I mentioned you can see a lot of different robots here very different structure they're they're they've got very different Form and Function but across each of the um pair of videos that you see here you'll notice that there's also a common pattern which is on the left side you tend to see something that's virtual and on the right side of the video pair you tend to see something that's physical and this illustrates the basic approach that my group takes to understanding how the body shapes the way we think which is to create AI that creates robots creates embodied AI so what do I mean by that uh what I mean is that in all of the projects that you see here um we create an AI that searches the space of all possible robots that could solve the tasks that we want the robots to solve um most people that are familiar with AI and Robotics and autonomous cars and drones have a rough understanding that AI is somehow optimizing or tuning the brain of the autonomous car or the Drone or the the robot what have you there's an underlying assumption in all of that current in in most current robotics which is that the AI Tunes the brain but does not tune the body Tesla cars are dreamed up mostly by humans and an AI Tunes their their brain or their control policy but of course nature doesn't work that way nature produced us and all the other intelligent organisms on this planet by carefully tuning body and brain simultaneously certain bodies facilitate certain kinds of behaviors and certain kinds of intelligent behavior and other bodies don't they obstruct that particular Behavior or that intelligent Behavior so in all of our work we ask the AI not just to tune the brain of a robot but its body simultaneously and as you can see visually here the AI often comes up with uh bodies that are well suited to whatever we want them to do so uh if you direct your attention to the very top left of the screen for a moment this is now a 22-year-old experiment but I think it's still visualizes the potential of this approach in this case we were interested in uh creating a robot that can brachiate that can swim swing across beams or tree branches or electrical wires for various inspection tasks and you'll notice that in this case the AI came up with a solution that in retrospect seems intuitive um the robot has to carry a very heavy battery which you can see in the physical robot in the top left there um the box that's at the bottom and the AI has figured out how to design the body of this robot so that it passive it's actually able to exploit the forward momentum of this heavy object this battery to facilitate its movement or brachiation across this physical beam so it's a simple example a simple robot a simple task but it demonstrates uh this interplay between AI robotics body and brain if the AI was not free to place the body uh to place the battery at a particular place on the robot's body it would be

much harder it would require more energy it would require a more complex brain for the robot to figure out how to move its heavy weight uh across this this be so that idea of tuning body and brain has suffused everything that we we work on um some other examples you can see in the top Center and the top right uh here we have a robot um that uh suffers damage its body changes over time so now the AI has to Grapple with not just designing a body but grappling with a body that changes one of the things that we as intelligent organisms here in the world and all embodied AI all autonomous cars all drones all robots have to deal with is entropy the world throws a lot of stuff at us we have to deal with wear and tear injury in our case we grow from a single cell into about 10 to the 12 cells we change massively in terms of our physical magnitude how do you continue to operate keep yourself alive and do whatever it is you need to do across massive morphological change that is not an easy thing to do and again it requires an AI if it's going to do this with robots it's got to figure out how to carefully tune body and brain to deal with the generation of behavior inside a body that is changing either unexpectedly due to injury and wear and tear uh or intentionally uh you can see in the very center panel um this is again a pretty good visualization of uh where designing body along with brain comes in handy um if we want to make flying machines or swimming machines we have to very very carefully tune the geometry and material properties of the body itself to realize flight um so what you're seeing in this middle panel here this is partway through the AI experimenting with the design of different kinds of wings for an ornithopter a drone that flies by flapping its wings this uh flexible Wing that you see here in the center of the screen this is a bit of a transition from traditional robots that are made of rigid structures like you can see in the top row into a more modern era in robotics which is sometimes referred to as Soft Robotics Material Science has come a long way in the last 20 years so we can now start to build robots embodied AI we can start to build uh robots out of materials other than rigid plastic and metals um and we can again we can start to move into an era in which robots like organisms can exploit the material properties of their bodies to facilitate whatever behaviors they need to do to survive or be useful to humans and so on so in the middle right panel um this is a highlight of some work that my group has done in collaboration with Rebecca Kramer Boo's Lab at Yale uh Rebecca's lab uh is famously advancing the state-of-the-art in Soft Robotics what can you get robots to do when they're made out of soft materials you can see an example of some of those soft robots middle right and a very different soft robot lower left which is exploiting its uh body properties in order to move in interesting ways one of the interesting things about Soft Robotics in my perspective is that it starts to usher in uh an era in

which robots can actually grow and complexify their bodies you can see these Hollow cubes in the middle right and these uh Hollow sort of uh Chambers in the bottom left expanding and Contracting as we push and pull air into and then out of the body of the robot suddenly now you have robots that can change their geometry they can change their volume they become what's known as thermodynamically open it's a fancy term for meaning that they can draw new material and new energy into themselves the thermodynamically open machines that you see middle right and lower left the only thing they're drawing into their body is more air but it's a start I mentioned already that humans grow from a single cell into 10 to the 12 cells every organism on this planet with a few exceptions starts small and gets bigger over its lifetime that fundamental morphological change start small start simple and gradually grow in size and complexity that provides scaffolding it provides a gradient on which to learn how to gradually grapple with the world around you most organisms again there are exceptions are not thrust into this world with all of their Machinery online from the beginning just the way I'm phrasing this is obviously int to to sort of dichotomize growing organisms and robots with fixed bodies autonomous cars are still very dangerous autonomous drones are still very dangerous to be around because 99.99% of the time they do the right thing but every once in a while they don't know what to do and no one knows what they're going to do in those uncertain circumstances that that is a very concerning situation as we start to now deploy robots and autonomous vehicles into everyday environments where they are in close proximity to humans why is it that in even with state-of-the-art AI and with all of Google's data centers and AI training algorithms we still can't rub out that 0.01% where no one knows what's going to happen part of the reason again is these machines are born with complex bodies we drop a controller into a one ton autonomous car made of metal and and plastics it's very dangerous we we don't grow autonomous cars from a very small simple lightweight machine that can't cause anyone any harm whatsoever and then when that simple small machine demonstrates and verifies to us that it's safe then we allow it to become larger and more complex it sounds like silly sci-fi why would we build a machine like that but again every organism starts simple and grows in complexity and if it doesn't do the right thing if it performs dangerous actions that are harmful to itself or fatal to Itself by definition it doesn't get any further that's again one of the ways of thinking about how the body shapes the way we think in my in my personal professional opinion any ma physical machines that we deploy into the real world they should start as very small lightweight machines that can't harm anyone they have a very limited number of actions that they can perform and they sort of cycle through all those actions and

verify everything and only then can they take more mass more energy into themselves can they recruit more material they can sort of be allowed to be thermodynamically open and grow and complexify themselves there's lots of ways in which we we're starting to create machines that grow and complexify themselves I just talked about these uh soft robots that can pull in air or or pass possibly fluids they could be hydrodynamic machines they could mechanically or magnetically connect to other robots that's Sort Of Swarm robotics that's another path to Growing machines at the moment most of these machines are still restricted to academic labs they also are not safe yet but I think in the long run they're going to be a safer alternative to dropping AI into very large complex heavy dangerous machines and crossing our fingers and hoping for the best okay so I've talked a little bit about rigid robots and soft robots um I want to try and talk as little as possible so there's some time for a good discussion but I I want to talk about what I see as sort of a third era of robot otics and embodied AI which is just starting to open up in the last few years which is biotics or creating biots and you can see two biobots uh on display at the bottom center of my slide here um a biobot is a robot that's made up of only biological components no technological components whatsoever so in the bottom left here uh krigman blackest and Len and myself in 2020 published a paper demonstrating the first biobot um this became known after publication in the popular media as xenobots x n o Zeno xenobots because these xenobots are built from about 2,000 frog cells and the cells were taken from a particular species of frog which is xenopus leis uh Michael Levan uh our biology colleague at T University uh is world-renowned for demonstrating that you can reconfigure genetically unmodified materials like for example frog cells and that rearrangement of living tissues not only does not kill the organism the organism is able to in some cases continue on doing what it does what it needs to do uh ingest materials survive reproduce in this reconfigured state there's a lot of biological implications for that one of the biological implications is that frog DNA does not code for frog what you're looking at in the bottom left the xenobot is about a millimeter in diameter so it looks like a speck of pepper to the UN to the unmagnified eye and yet it's able to walk around the bottom of a petri dish it's not quite doesn't have all the features of a living organism but it's got enough of them that it's motile it's able to get itself from point A to point B so one of the one of the the other implications for AI of this biological discovery that you can rearrange genetically unmodified living tissues is that maybe we can task an AI with discovering novel rearrangements of living tissue to produce robots to produce something that moves around and does something useful on a human's behalf so that that's what I mean by a biobot biobot that's made from in this case genetically

unmodified cells um the the swarm of xenobots that you see in the bottom uh right uh as you can see they're sort of pushing around some material in their their dish this um sort of visually hints at applications for this this type of robotic technology which is they might be able to act like very very small Roomba uh Robot vacuum cleaners in the future they might be able to collect microplastics out of waterways or cancer cells out of bloodstreams the the Swarm that you see that's cleaning up uh in this slide at the bottom right the material that they're cleaning up is actually other frog cells it turns out that if the AI designs this swarm just right and the Swarm that you're looking at this is an AI designed swarm the SW the AI came up with the shape for each member of the Swarm the Swarm is pushing these little white circles these which are individual frog cells pushing them into piles turns out these frog individual frog skin cells at a certain stage of development are sticky and they Clump together into a pile and some of these piles if they're big enough if they contain enough frog cells they will grow very small hairs on the surface cells the cells that are on the surface of the pile and those uh those little small hairs are called cilia they're usually used to pull uh dirt and pathogens off the skin off the body of of frogs adult frogs here when those cilia grow on small piles of frog cells they're able to uh exert enough Force against the surrounding water that these piles start to move and so what you have in essence is a child xenobot this swarm wor pushes cells together and in essence makes copies of themselves this is another implication of this work is that in this case the AI has figured out how to design robots that replicate they make copies of themselves by by finding raw material in their environment and constructing copies of themselves this has been a long-standing dream in robotics dating back to um dating back a very long time to John Von noyman in the 19 50s who had a thought experiment it would be great if we could create robots that would create copies of themselves which would create copies of themselves and if those robots do useful work for humans as a side effect for Von noyman that was creating moon bases and then Mars bases and then colonizing the Galaxy which sounds great um but but the the seed of this idea is if we want uh robots to really be useful at scale instead of manually constructing billions of robots and then deploying them to do something useful which is expensive would be much cheaper to make one robot that does something useful for us and by the way it also makes two copies of itself which does more useful work for us and four and 8 and 16 and so on we're not there yet with the xenobots but it's a demonstration that that is possible and again all of that becomes possible because the AI is designing both both the bodies and the brains of these robots this is very far now from the traditional view of AI and Robotics where we build a robot body we humans build a robot body and then the AI

Tinkers with the brain of that fixed machine so part of the reason why I'm here today and part of the message of my group is we need to think more broadly about how to combine AI and Robotics and possibly synthetic biology and when we do when we think more broadly there are Whole New Paths that open up to ways in which we might create in the future not yet but in the future create intelligent useful and safe machines in the current ERA in AI there is one particular approach which is autoc completion of tokens which has come to dominate the field and and come to dominate the popular imagination we all kind of have an understanding more or less of what chat GPT is doing and there are some very strong lobbying organizations out there that are bent on convincing us that if we just do this with more compute more data we will eventually get to Safe machines my contention is there just isn't enough data out there to make non-embodied AI like chat GPT and stable diffusion and all the rest to make them safe we have to think differently about designing bodies and brains of machines simultaneously to realize this long-term goal okay um I've been talking for a while um I'm going to stop and I'm happy to take questions or uh engage in some discussion and I'm happy to come back to any of these experiments and provide more detail if that's helpful over to you Daniel thank you wow awesome what a cornucopia of bodies and Minds it was a great overview I was really struck by some of the similarities and the convergence on whole of life cycle design and kind of holistic Design coming from on one hand a systems engineering and a materials perspective and on the other hand from the biology perspective with like Eco evoo and this convergence upon needing to think about how the endtoend function maybe even past the point of functionality like into the planned graceful decay of a robot as well so it it brings in a lot of topics that even from an outsider's perspective seem to be put as kind of secondary so that's very cool okay great I'm looking forward to what people in the live chat write my first question is how over these uh 22 years how have the materials the theories like the contexts advances and Turing computation all these kinds of things how have they intersected and just what has the ride been like as you pursued these questions yeah I think it's um the short answer again is is focusing on the physical aspect of AI and Robotics so the materials from which we can build machines has changed over 20 years and from from my perspective the the experiment the top left there that was something I did as part of my PhD you know the materials at the time it was it was very hard to build a robot you got you bought some Motors you bought a battery you bought some metal you bought some wires and you wired everything up there was there was the assumption that bodies were fixed and not only that but they were difficult to make so once you made one you were very careful with it to make sure it didn't

change that it didn't become damaged and that's seemed to comport with a lot of the theory in AI and Neuroscience which which had the same sort of idea that that the brain or in the case of Robotics the control policy is the Puppeteer it's something that's pulling the strings of a fixed thing either the body of an organism or a robot and if you look at a lot of a lot of theory in both Fields AI and Neuroscience that that assumption runs so deep so for example an active inference or you know uh the free energy principle we want to reduce surprisal that there's a fixed set of actions that we perform to try and reduce the surprise between what we're sensing and what we predicted we would sense given the past action where do those actions come from why are they fixed does the set of actions grow over time maybe the the sensory data that's coming is coming from a new sensor that's just coming online or a sensor that's recently duplicated now there's two of them but they're not quite reporting exactly the same thing there there's a whole bunch of assumptions underlying a lot of the theory about active inference uh predictive coding you name it you pick your concept from Neuroscience or cognitive science or AI once you once you peel back those assumptions imagine the robot's body changes imagine the robot splits in two and becomes two copies of itself you know a lot of the the theory and the formal the the formal underpinnings of that theory break down you start to get into ill-posed questions which force you to now think about how do you fundamentally change the theory if you have a hierarchy of actions like in predictive coding or active inference what if that hierarchy is growing and changing over itself is growing and changing over time how do we address that in a formal manner so to get back to your question I think these advances in what we can do physically we can build robots now out of soft materials we can build robots out of living materials which on their own will grow and divide and seek out energy and material on their own that th those physical machines these odd new kinds of creatures are militating they're pushing against the theory and specifically they're pushing against these unspoken assumptions that lie underneath a lot of this theory about what's required to act intelligently in a complex world that's awesome like the real world and the territory expanding into our unknown unknown okay there's a bunch of questions in the live chat so okay I'm just going to go for them give any answer that you like okay sure David Williams wrote how do you think about the controllers in your robotics embedded AI at least today is rather hard batteries and chips pcbs not soft and not easily synthesized locally so how do you think about the controllers in your robotics yeah great question so right exactly the controllers are dealing with hard rigid fixed components we need we need to start thinking about controllers that can in which for example the input layer and the output layer can grow

and Shrink over time there may be new sensors or new input coordinates that are growing or being attached to a machine and the controller needs to be able to carefully deal with those new input channels while the machine is operating same thing goes for the output Channel there may be new actions or New Dimensions of action along which the machine uh can act and control policies reinforcement learning uh all the rest of it does not all those assumptions that make reinforcement learning work which is what drives most autonomous vehicles at the moment assumes that the dimensionality of input and the dimensionality of output the things that the machine can do and sense are fixed during training or during Behavior generation during execution that is absolutely not true in any organism on the planet and that's becoming increasingly untrue for for our our coming machines now how to do it well I I don't have any answers but we have to figure it out that's you were asking a question about thinking about controllers that's a a concrete example about how we have to rethink control policy optimization even if we're not thinking directly about the body even if we just focus on the control policy and ask what happens as the input and output channel the dimensionalities of the input and output channels change during Behavior execution yeah just one short point on that it's like training with a fixed set of perceptual elements or of affordances or of actuators it's like training on one point in a larger space of the adjacent possible of like bodies or of architectures so then okay we're bringing all this compute to train a special case in the fixed setting and that's not even how the smallest organism works so that just again like kind of shows that point okay I ju sorry before we move on from that point I just to again illustrate how the body shapes the way we think in the case of a growing biological body there are new input channels that come online you know throughout our life time but they don't appear denovo you know whatever it is whatever that new input channel is as as we're growing we just have more you know more s sensory more sense cells the signals that they're sending into the peripheral and central nervous system are not orthogonal to whatever else is already coming in as input because new input channels or new cells are slowly dividing and at the moment of division they're providing exactly the same signal as some other sensory channel that already exists so the body or in this case biological growth provides an immediate scaffold a gradient in in robotics it's it can be very scary to think about like attaching a sensor to an autonomous vehicle what the hell does it do with this new information that's coming in because we haven't thought carefully about how to add that new sense modality to the machine again we have to look to Nature that every new sense sense modality is gradually coming online and gradually drifts away or becomes increasingly orthogonal to the starting input

modality so that's how we should if we did that physically with machines it would simplify reinforcement learning or would make it easier for reinforcement learning or what have you sorry let me re share my screen here it would simply make it easier for the uh sorry something seems to have gone wrong here give me a moment yep okay all right yeah it what makes things easier on the control policy optimization process if new sense organs and new motor outputs are coming online but they resemble things that already exist that's super interesting um brings up a lot of questions about like self and on self recognition and what is a self as new and different senses and actions come online sure um okay Pros kavi asks do these biobots have any sense of agency what is your sense I'm quite intrigued by the idea that Beyond a critical point they start growing hair and do these biobots act independently of each other and also what happens at a group level so what's your sense of agency in bio Bots and I guess the biobot and the group level yeah it's a great question so I'll start with a disclaimer I'm not a biologist I'm a computer Scientist by formal uh training so I I can only say so much about what the cells are doing and what they want to do um I I definitely uh I I definitely follow in the footsteps of the late Daniel Dennett in that when we talk about agency we all each of us individually has to decide whether or not we take the intentional stance or not it's in my opinion it's a point of view if it's easier to explain what the xenobots are doing by talking about what the cells want to do like grow cyia and coordinate their actions fine if it's easier to explain what the xenobots are doing by not taking the intentional stance and describing cells as mechanical components that are transforming input into out into actions that's fine too it's uh this is something also that comes from my colleague Mike Levan it depends if as scientists if we want to try and explain and understand what these machines are doing if taking the intentional stance makes explanation easier fine if not then not but attributing agency is sort of an objective property of the Bots or the cells themselves independent of us is observers to me that's philosophically and practically uh problematic as far as I know there is no objective measure of agency in cells let alone inorganic robots super interesting that's like the second order cybernetics or the Observer Theory or the poly Computing question which is to say just looking at something and then treating one's perspective um objectively the case it is objectively the subjective experience absolutely now that being said again there is there is an empirical side to this we can make some progress in understanding the xenobots uh by comparing them against a control so instead of cells if these were you know magnets or some complex mechanical system in which more of us are comfortable in saying there is no agency it's just a you know it's a bucket of cogs doing something and that control does not exhibit

kinematic self-replication for example or it's much harder for the AI to figure out how to put together non agential components to do what it is then then it I feel a little bit more comfortable by saying the cells are doing something more now I don't know whether it's agential or they want to do something or if it's free will or Consciousness I don't know but if but if we can point at biobots or machines that are built from biological components and say it's easier to get them to do things because they start to um they become complicit in the overall goal compared to Mechanical Parts which don't okay and again as a roboticist in the in the top the top and the middle rows that you see on my slide here when we do build things out of metal and rubber and Plastics and Ceramics it's usually super hard it's really hard to get them to do whatever we want them to do we've been working on robotics since the end of the second world war and we've got the Roomba and you know maybe we've got autonomous cars we're getting there it's taking a really long time because robotics is really hard it's really hard to convince physical materials to to adapt and do something useful and safe on the flip side we've been working on Xeno Bots at the bottom here we've been working on them for about five or six years and we've got roombas we we're making faster progress in robotics when we build from cells then when we build from metal suggests you know the cells are somehow helping I don't know that they want to help got to be careful there that's the intentional stance but when you deal with smart when you when you try and compose machines from Smart machines and cells are smart machines we're making f i I know I'm biased but I think we're making faster progress than when we build machines out of inert materials yeah super interesting um okay I'll read some comments from sure David Clement David wrote does your work incorporate agential hierarchies for example the xenobots grow by replicating the initial seed cell into a higher order system and is it critical for lower order systems to act as a component of a virtual machine meaning that they have a Target behavior that is less than the higher order system and that's kind of related to to precautionous question as well like how do you bridge that from the individual component into the Swarm or the aggregate yeah it's a really it's a really good question so absolutely I think the the when we started working on the xenobots and Mike Mike Levens started to talk about machines made of machines made of machines that definitely has influenced the work in my group to focus on this issue of hierarchy I don't know about agential hierarchy again we just talked about agential agency that's maybe a a subjective stance but we but definitely you know why would you want to build machines out of machines out of machines at the moment our state-of-the-art robots like autonomous vehicles are not hierarchical they the control policy operates at the level of the uh

of the machine as a whole for example if there's an emergency blowout of the tire in an autonomous vehicle the tire itself the rubber that makes up the tire doesn't deform and try and fix or reduce surprisal locally it can't it's rubber it's inert material we don't we don't have machines built of machines built of machines yet but as biology in general and the noots in particular demonstrate there's an Adaptive advantage to being a hierarchy of physical things of physical machines if there is a surprising event at the level of the machine as a whole but that surprise trickles down through the hierarchy it's unlikely that everyone at every level of the hierarchy is going to be surprised someone somewhere in the hierarchy is going to say from my local View at least on this bigger surprising issue I know what to do so let me start to communicate to my peers and up the hierarchy to deal with surprise that would be from an engineering point of view that would be a good thing to have in you know big heavy fast-moving robots that are near humans you there's always going to be some surprising event that the vehicle as a whole has never seen before there's great YouTube videos of you know horrifyingly you know scary surprising edge cases for autonomous vehicles okay we're never going to fix every Edge case what we can fix is to make hierarchies and maybe agential hierarchies where local surprise can be handled or or Global surprise can be broken down into local surprising events which can be handled locally um if I understood the second part of your question is how do we design that hierarchy should the small Parts be trying to pull in the same direction or be trying to solve some part the goal of the higher level I think that's a super interesting question and I don't think that the answer is obvious it may be that smaller parts pursuing orthogonal goals ends up actually being useful uh may end up being useful just to give you a quick example um if you want if there's a surprising event and you've got a whole bunch of semi-independent machines organized in a hierarchy I would argue that every single one of those members of the hierarchy should have a slightly different body and brain it should have a slightly different Form and Function you don't want a monoculture you don't want all the parts being smaller versions of the bigger parts and trying to achieve smaller versions of the same goals because then you've got you've basically got a committee in which everybody thinks and feels the same way and as we know from humans that's a dangerous thing right you get group think or group act you actually want a hetero culture you want a whole bunch of things that are unique in terms of Form and Function and that maximizes your chances that someone somewhere in the hierarchy says just because of the way I'm built and the way I think with my local control policy I know what's going on and I have the seed of a solution here's the seed you all figure out what you need to do to make it a reality at the larger

level that's another aspect of where the body comes into play yeah thank you like everywhere is the last mile from somewhere things have to be addressed locally no matter how you think about a Communications architecture distally everything and embodiment calls our attention back to that like it has to be somewhere locally so then why not take that as the starting point instead of like this kind of resource Challenge and then about like the multiple subunits when there's a uh damage to the nest of an ant colony or there's some things spilled on the surface it's like not that every single nestmate makes a perfect Pebble move it's that 51% accuracy with a bunch of nonspecific flurrying of activity just like kind of stress or these more generic higher order in that is what allows nestmates with different brains and bodies to fulfill their own Paths of least action and then colonies for which that doesn't clean up the mess or it cleans up too well and there's externalities those colonies are swept off the table and then we see the Persistence of collective systems that could figure that out in their growth from a little colony to a big Colony absolutely great great example um i a question you you mentioned uh both safety as well as like reliability and how do you think about capacities and evaluations on diverse intelligences we're all familiar with ram CPU hard drive storage some of the Von noyman type architectural descriptors however how do we even think about what does that rubric or report card even start to look like when we we know that there's complex interactions with the niche and when the kinds of capacities that we're talking about may have even open-endedness yeah great great Point great point right so you know we we are the beneficiaries of you know two big revolutions one of them is the AI Revolution but then the older one is the digital electronics Revolution right so Digital Electronics works we all have you know a supercomputer in our pocket like there's no arguing with it it's an incredibly powerful way to make machines that internally communicate quickly and richly and then can communicate with other machines I mean that's it that's that's the that's the information age that we're in it's been so successful that it's hard to think about Alternatives or why we would even bother thinking about Alternatives but again living systems a lot of what cells do they rely on electrical communication but they also rely on mechanical communication chemical communication uh thermal communication cells are using all mod physical modalities not all physical modalities but as many as they can get their hands on simultaneously all the time why why don't they just abandon everything and do everything purely electrically like our modern civilization has done because it's dangerous you don't have a diversified portfolio so uh one panel here that I haven't talked about is the one in the bottom right which is basically just uh what you're looking at is what's called a granular material it's a material that's made up of a

bunch of grains in this case the blue circles that you see these are little just rubber rubber pucks and there is an oscillation being supplied at the left hand side and you can see that this leads to interesting non uh nonlinear vibrational Behavior within this material what does that have to do with robotics or AI if you view the vibrations as the carrier of information so if a puck is vibrating that's a one if the puck is not vibrating that's a zero now you can start to imagine creating materials that communicate Shannon information throughout the physical structure not with electricity but with a different uh a different modality Dynamics or vibration and it turns out that can actually compose these metam materials to embody logic gates if you vibrate one particle or another particle but not both and not neither you can see watch a third particle and it will either vibrate or not in accordance with an exclusive ore gate and you can build this up now again why would you do that we can make an exor gate that's vanishingly small and vanishingly fast in digital electronics why would you ever want to do something different because it turns out there are advantages of communicating with vibration rather than electricity under certain conditions having a machine that can communicate between distant parts of its body through mechanical vibration as well as electricity has an advantage over a machine that can only communicate long distances within its body electrically I won't go into the reasons but you can intuitively start to understand that so again I think we need to if we if we're serious about creating safe and useful autonomous machines we have to break out of the digital electronics assumption that that's the only way to do things we have to break out of the assumption that non-embodied cognition is the way to go and it's easy to just drop it into a physical body and we're good to go we have to peel back some of these very deep assumptions about the right way to do things that have built up in our society since the second world War because a lot of those technologies have been very successful nothing wrong with them but when we come to apply them to creating safe and useful machines not always the right thing or the only way to approach things that's really interesting it's like a sort of generalized compute concept where we could talk about well these are the chemicals that it can detect with this Fidelity and here's its tactile interface here's its electrom magnetic capacity for sending and receiving that's what kind of motivates or complement the generic Theory like free energy principle which doesn't tell us about how anything is in particular but then sets us up with kind of the framework to plug in these different modules and then it's an empirical question and then right here is sort of um virtual body and a real body in that's also very interesting H how how does that work like in a collaboration or with a graduate student like how do you bounce this digital adjacent possible off of the material and the more costly um

implementation with embodiment yeah well with so with grad students and postdocs or whoever you know I'm collaborating with that's kind of starting out you know this can be a very frightening Prospect for someone who's trying to get into Ai and Robotics because it looks like everything's been done it's solved like we all we got you know we just have to wait for Google and Microsoft to buy more compute and data and they'll they're going to finish off the last 1% of the dangerous Behavior so if you're trying to contribute to society's goal of making useful and autonomous safe machines as starting out what do you do it it looks like this massive brick wall there's there's no entry point so my take on this is again is that we may be going about this all wrong right the assumption that electricity should be the carrier of information inside an autonomous machine that's an assumption why electricity why not vibration why not something else so even if you start to think about the Alternatives the immediate reaction is well it's it's not going to be as good maybe maybe but if you think about vibration you were just mentioning like compute we can use vibration for compute but vibration is movement so the minute you start to think about vibration as the carrier of Shannon information you're now conflating action with computation they cannot be separated dayart convinced the West 400 years ago that they're separate they just are and you you look at Ai and Robotics what a surprise these you know the our attempts to create you know AGI is by Camal there's one team that says it's going to happen in computers and the other side that says it's going to happen in physical machines that's the cartisian Legacy that they're separate but the minute you look at some very humble material like the one in the bottom right it's a bunch it's 12 hockey pucks next to one another there's no cartisian division anymore between body and brain there there is a body and there is a brain there but it looks very different from anything we we would usually consider and it's there's no there's no value judgment here it's not better or worse maybe it is depending on what your metric is it's just very different and so with grad students and posts I encourage them to pursue that what you know could we do things completely differently and in the long run might that be a better way to do things what who knows we'll see awesome I I'll make one comment and then ask a last question you brought up Dart and and that's the r extensa r cognant uh dualism between the the the the thinking and the non-thinking substance and embodied cognition embodied intelligence provides both a an operational instrumental and an ontological counterargument or complementary perspective which is just well in practice and in actuality take it or leave it they are inseparable and so so at the very least that that starts to ratchet and LeapFrog the discussion about what is mind and body and uh you know you started with with pointing out how

important it was to co-evolve in the complementarity of mind and body and it's like they're two separate things that need to be complimentary and tangling and also maybe they're integrated and blurred in even deeper ways than the dance so that's like it's it's an empirical entry point into what otherwise is a thought experiments which can have utility but also can be just arbitrarily misleading absolutely one of my one of my former mentors Inman Harvey at the University of Sussex used to talk about robotics as philosophy with a screwdriver you know it's not just armchair philosophy it's when you start to build some of these machines maybe in retrospect you you in in the case of the metam materials project for me in retrospect I said oh my god there you know mo action and cognition are not complimentary separate things that are complimentary they're one and the same thing it's not embodied cognition it's not an adjective of a noun it's embodiment is cognition there they're not there are no they're not there are not two things here there's just one thing very hard to think about it's so alien to a western mind but you know it just is sometimes I think about that in terms of like adjectives getting added in front of a word and then the term expanding and then it it just encompassing oh of course cognition is ecological embodied inacted etc etc etc etc and then so it kind of like needs to be distinguished and then it subsumes again and that's part of um in closing like what are you excited about where can people continue to learn more what would you say to a person who's wanting to like go in this area yeah um okay great question so Google my name and it'll take you into lectures and papers and tutorials and and people want folks want to email me that's perfectly fine again Google my name you can find my email happy to put you in touch with the right people um I think it's you know it's easier than ever to get started you can go to chat GPT and say you know create some tutorials for me to start coding up robots you know they're I ironically non-embodied AI provides a good on-ramp now not just for reading about these ideas or listening to people talk about these ideas you can start coding them up in a way that was easier than ever that's easier than ever the old days you know you had to learn C and then you know go on from there very easy to get your hands dirty maybe not with physical materials but you can create like you see in the left of each of these panels you can create machines that are virtual they're not physical but they're embodied it's another Point that's important to make embodiment does not imply physicality you need to be able to push against the world and observe how the world pushes back but the world and you may be virtual like you see on the left or physical like you see on the right so you can actually relatively quickly get your hands dirty with playing around with embodied AI uh these days and I encourage everyone to do so cool any last comments I I would just say um I was at the uh computer

vision and pattern recognition conference cvpr a few weeks back this is one of the flagship AI conferences 15,000 attendees and after my talk a lot of grad students came up and they said listen I you know you sort of demonstrated there's another path here that I was feeling depressed or anxious about how to make progress in AI when these goliaths are you know have these data centers at their beck and call I would just uh encourage everyone that when you think differently about all this stuff there are New Paths that open up they may not in the long run be the right path but there are alternatives to this monolithic predict the next token idea which is currently you know invogue it's it may be the beginning of the end um but I think this is just the end of the beginning we've we've figured a few things out there are some things that work but they're still producing not quite useful and definitely dangerous machines there is room for improvement and there's nothing that says that only Google Google with all its resources is going to be the one that can figure out these improvements think differently try some of these uh alternative approaches and maybe you will be the one um you know uh that comes up with the answer whatever it is cool good luck awesome thank you Josh really appreciate it thanks for having me yeah and uh till next time thank you bye e